

T-CELLS

Regulatory T cells promote innate inflammation after skin barrier breach via TGF- β activation

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Regulatory T cells (T_{regs}) use multiple mechanisms to attenuate inflammation and prevent autoimmunity. T_{regs} residing in peripheral (i.e., nonlymphoid) tissues have specialized functions; specifically, skin T_{regs} promote wound healing, suppress dermal fibrosis, facilitate epidermal regeneration, and augment hair follicle cycling. Here, we demonstrated that skin T_{regs} were transcriptionally attuned to interact with their tissue environment through increased expression of integrin and TGF- β pathway genes that influence epithelial cell biology. We identified a molecular pathway where skin T_{regs} license keratinocytes to promote innate inflammation after skin barrier breach. Using a single-cell discovery approach, we identified preferential expression of the integrin $\alpha\beta 8$ on skin T_{regs}. Upon skin injury, T_{regs} used this integrin to activate latent TGF- β , which acted directly on epithelial cells to promote CXCL5 production and neutrophil recruitment. Induction of this circuit delayed epidermal regeneration but provided protection from *Staphylococcus aureus* infection across a compromised barrier. Thus, $\alpha\beta 8$ -expressing T_{regs} in the skin, somewhat paradoxical to their canonical immunosuppressive functions, facilitated inflammation acutely after loss of barrier integrity to promote host defense against infection.

INTRODUCTION

Barrier tissues such as the gastrointestinal tract, lungs, and skin are home to relatively large proportions of regulatory T cells (T_{regs}) in both rodents and humans (1, 2). T_{regs} seed these organs early in life and receive local cues that facilitate their survival and function (1, 2). Skin is in constant contact with the outside environment and highly prone to barrier breach in the contexts of both superficial abrasion and full-thickness wounding. Thus, regulating immune responses in this organ is important to avoid chronic inflammation incited by environmental stimuli. However, these mechanisms must be precisely balanced to control and eliminate potential pathogens. The cellular and molecular mechanisms used by T_{regs} that establish and maintain this balance are largely unknown.

Given the specialized function of tissue T_{regs} (1–3), we hypothesized that these cells might exhibit a transcriptional and proteomic profile that is imprinted by their environment. Single-cell RNA sequencing (scRNA-seq) studies give credence to this hypothesis by revealing heterogeneity among T_{regs} present in various lymphoid and nonlymphoid tissues and by identifying transcriptionally related cell subsets primed for a nonlymphoid environment (4, 5). Nonetheless, whether T_{regs} in the skin are specifically adapted for tissue-localized, cell-cell communication remains unclear.

Transforming growth factor- β (TGF- β) signaling is a compelling candidate pathway to mediate tissue-specific T_{reg} functions because this cytokine plays a major role in tissue repair/remodeling after injury (6–9). Furthermore, there is an intimate association between TGF- β biology and T_{regs}. This cytokine acts on naive CD4⁺ T cells to help differentiate them toward T_{regs}, leading to regulation of various immune responses, especially T helper 17 (T_H17) inflammation

(10–13). In addition, T_{regs} express the receptors for TGF- β , and they can secrete large amounts of TGF- β and express integrins capable of activating latent TGF- β (14–16). Despite numerous studies linking TGF- β and T_{regs} and the role these play in tissue homeostasis, it is unclear whether and how T_{regs} use the TGF- β pathway in the context of tissue repair.

Here, we found that T_{regs} residing in the skin had a tissue-specific transcriptional profile that potentiates cell-cell communication with skin epithelial cells. T_{regs} in close proximity to keratinocytes were enriched for interacting receptor or ligand gene partners, including *Notch1/Jag1*, *Egfr/Areg*, and *Tgfb1/Tgfb2*. TGF- β pathway genes, especially the TGF- β -activating integrin $\alpha\beta 8$, were among the most preferentially expressed in skin T_{regs}. Expression of $\alpha\beta 8$, but not TGF- β itself or TGF- β receptor (TGF- β R), endowed these cells with an ability to directly promote innate inflammation after epidermal injury. Our results suggest that T_{regs} in close proximity to skin epithelium maintain tissue integrity by balancing reparative and proinflammatory functions.

RESULTS

T_{regs} in the skin were transcriptionally enriched for TGF- β and integrin pathways

To begin to understand the transcriptional heterogeneity of skin-resident T_{regs}, we performed scRNA-seq on cells isolated from the skin and skin draining lymph nodes (SDLNs) of forkhead box P3 (Foxp3)-green fluorescent protein (GFP) reporter mice (Fig. 1A and fig. S1A) (17). Comparative analysis with PANTHER (18) revealed a set of transcriptional pathways preferentially enriched in skin T_{regs} over T_{regs} from SDLNs. Both TGF- β and integrin signaling were preferentially enriched in skin T_{regs}, with greater than a threefold enrichment in both (Fig. 1B). We performed gene set enrichment analysis (GSEA) (19) to verify increased expression of these pathways in skin T_{regs}. Both TGF- β and integrin-associated genes were significantly elevated in these cells (Fig. 1, C and D). At an individual gene level, expression

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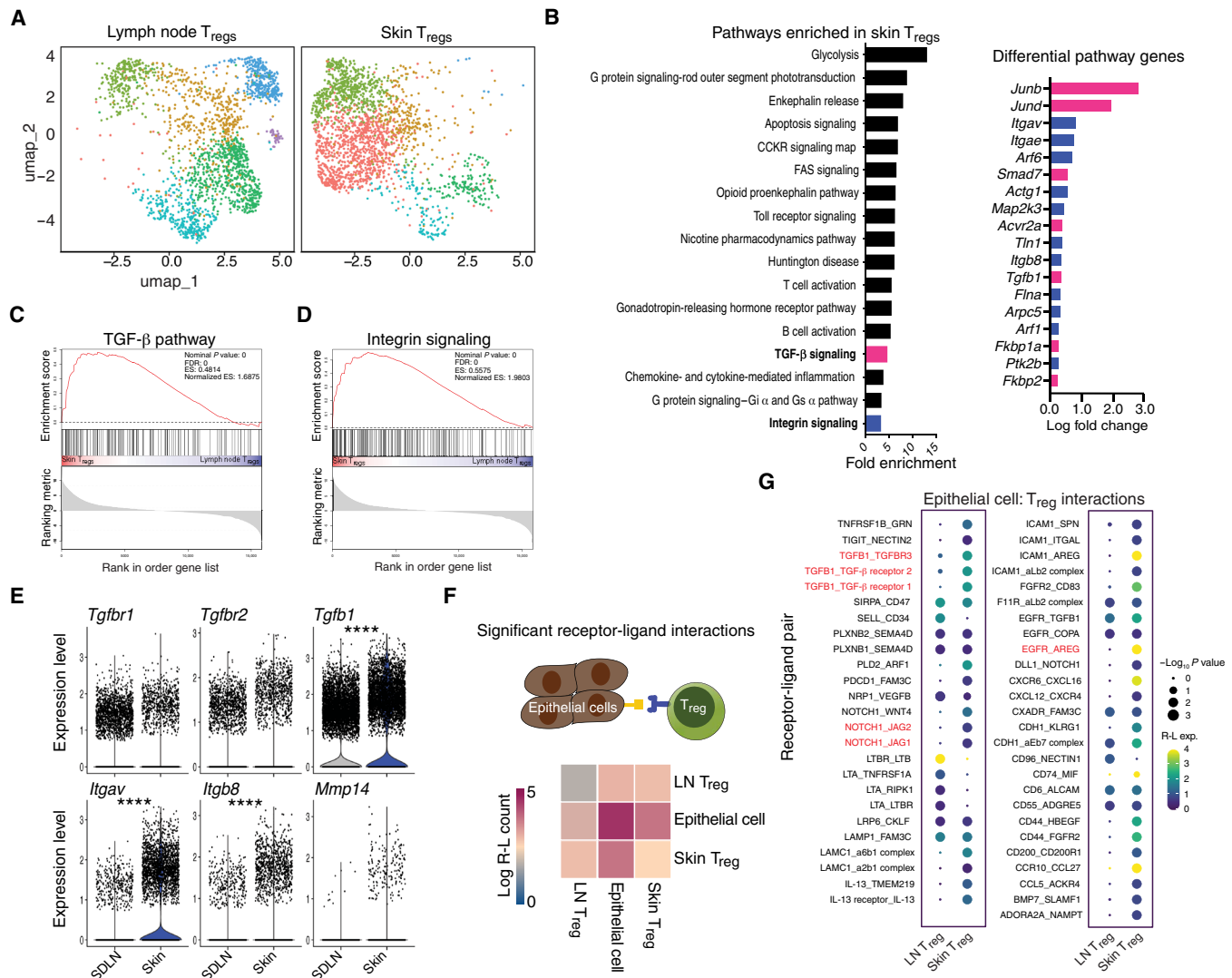


Fig. 1. T_{regs} in the skin were transcriptionally attuned for their local tissue environment. (A) UMAP of unsupervised clustering of scRNA-seq data comparing skin and lymph node Foxp3⁺ T cells isolated from Foxp3-GFP mice. Each tissue represents cells pooled from three to four mice. (B) Significantly enriched [false discovery rate (FDR), <0.05; Fisher's exact test] PANTHER gene ontology pathways identified from genes differentially expressed (nonparametric Wilcoxon rank sum test) between lymph node and skin T_{regs} in single-cell sequencing data. (C and D) GSEA showing enrichment of genes for skin versus lymph node T_{regs} probing TGF-β (C) or integrin (D) gene sets. (E) Differential gene expression analysis for select TGF-β pathway genes. ES, enrichment score. (F) CellPhoneDB receptor-ligand interaction repository analysis identifying significant ($P < 0.05$) interactions between lymph node (LN) or skin T_{regs} from scRNA-seq data (A) and epithelial cells (21). The heatmap indicates number of significant interactions between two cell populations. (G) Overview of statistically significant interactions between epithelial cells and T_{regs} using the CellPhoneDB pipeline. Dot size indicates interaction P values, and color indicates expression means of the receptor-ligand pairs between two cell populations. scRNA-seq data are derived from a single experiment where either skin or lymph node T_{regs} were pooled from three to four mice. For differential gene expression Seurat was used to perform a nonparametric Wilcoxon rank sum test. **** $P < 0.0001$.

of the TGF-βR constituents, *Tgfbr1* and *Tgfbr2*, was not increased in skin T_{regs}, whereas *Tgfb1* was significantly elevated (Fig. 1E). In addition, both *Itgav* and *Itgb8*, which form the TGF-β-activating integrin αvβ8 (9), were highly enriched in skin T_{regs} (Fig. 1E). *Mmp14*, which can aid αvβ8-mediated activation of TGF-β, trended toward significant enrichment (Fig. 1E) (20). *Itgb6*, another TGF-β-activating integrin, was expressed at very low levels on skin T_{regs} (fig. S1B). Therefore, comparative transcriptional profiling of T_{regs} revealed a distinctive skin T_{reg} signature defined by enrichment of TGF-β and TGF-β-activating genes.

T_{regs} in the skin were preferentially poised to interact with epithelium

Given the unique transcriptional signature of skin T_{regs} and their previously described tissue-specific functions (1–3), we hypothesized that enriched pathways would include genes functionally relevant to skin epithelium. To directly test this, we screened our skin and lymph node T_{reg} scRNA-seq data against a published scRNA-seq dataset of murine epithelial cells using the CellPhoneDB repository of receptor-ligand interactions (21, 22). This analysis revealed that skin-derived T_{regs} had more than double the number of bidirectionally

significant receptor-ligand interaction pairs than T_{regs} from the SDLNs (Fig. 1, F and G). Among significant T_{reg} -epithelial cell interactions, 31 were specific to skin T_{regs} . Significant interactions included gene pathways identified to be involved in tissue-specific T_{reg} function, including *Notch-Jag* and *Egfr-Areg* (23, 24). *Tgfb1-Tgfb1* interactions were also preferentially enriched in skin T_{regs} (Fig. 1G).

We have previously observed that T_{regs} in the skin primarily localize to hair follicles where they facilitate epithelial stem cell differentiation through *Notch-Jag1* interactions (23, 25). Nonetheless, specific analysis of the skin from Foxp3-GFP mice revealed that a small fraction of Foxp3⁺ cells localize to other anatomically distinct portions of the skin, mainly the interfollicular epidermis (IFE) (Fig. 2A). This region forms the functional skin barrier and is preferentially

inhabited by dendritic epidermal T cells (DETCs) expressing a restricted $\gamma\delta$ T cell receptor (TCR $\gamma\delta$) (26). To further elucidate skin T_{reg} epithelial interactions, we used an established enzymatic approach to separate skin epithelium from dermal layers (27). This method successfully isolated epithelial localized cells, as demonstrated by enrichment of DETCs (fig. S1C). T_{regs} were readily identified in the epithelial fraction by flow cytometry, albeit at significantly reduced frequencies when compared with the dermis (Fig. 2B).

We hypothesized that T_{regs} associated with the epithelial skin niche represent a distinct transcriptional subset. To test this, we isolated T_{regs} directly from dermal or epithelial fractions and performed scRNA-seq on both groups. Contrary to our hypothesis, unsupervised clustering analysis and sample integration revealed little difference between the two cell populations, with the epithelial

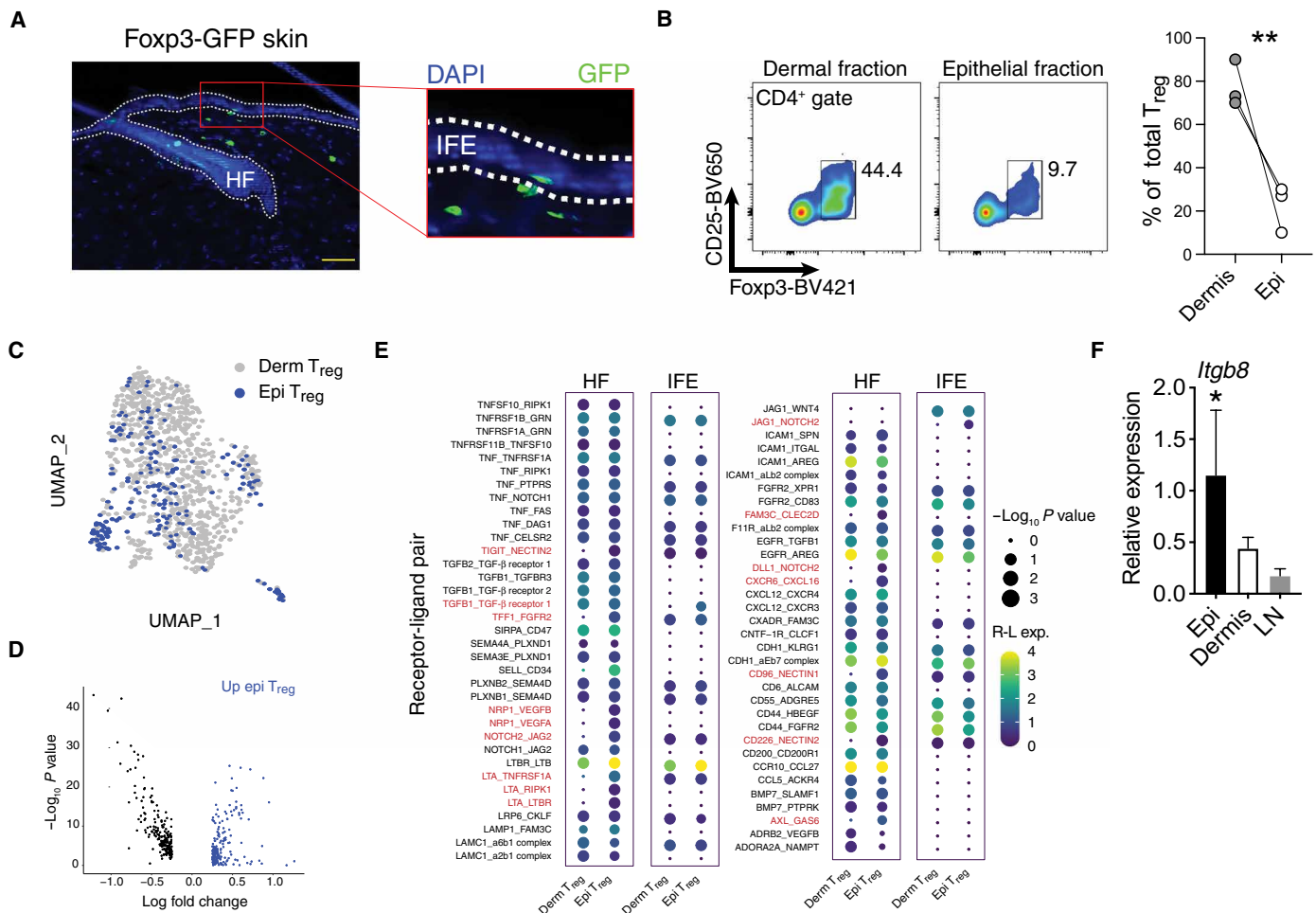


Fig. 2. Both dermal and epidermal T_{regs} were poised to interact with skin epithelium. (A) Representative immunofluorescence imaging of skin sections from Foxp3-GFP mice and counterstained with DAPI. HF, hair follicle. Scale bar, 200 μm . (B) Flow plots and quantification of Foxp3⁺ cells in the epithelial and dermal skin fractions. Lines indicated matched samples from the same skin sample. (C) UMAP of unsupervised clustering and integration of scRNA-seq data from dermal and epithelial derived T_{regs} isolated from Foxp3-GFP mice. Each sample is pooled from four mice. (D) Identified differentially expressed genes between dermal and epithelial T_{regs} calculated from nonintegrated scRNA-seq data. (E) CellPhoneDB identified significant ($P < 0.05$) receptor-ligand interactions between dermal node or epithelial T_{regs} from scRNA-seq data in (C) and epithelial cells from reference (27). Epithelial cells were further stratified into hair follicle or IFE cells on the basis of *Krt14*, *Mt2*, *Cd34*, *Ptn*, and *Postn* expression (see fig. S1). Dot size indicates interaction P values, and color indicates expression means of the receptor-ligand pairs between two cell populations. (F) Quantitative RT-PCR for *Itgb8* expression on T_{regs} isolated from epidermis, dermis, or SDLN and normalized to *Hprt*. All data are representative of at least three independent experiments, and error bars represent SDs; $n = 3$ to 4 mice per group. For comparison between two groups, Student's two-tailed t test was used. For comparison between multiple groups, one-way ANOVA was used. * $P < 0.05$ and ** $P < 0.01$.

fraction of T_{reg} s homogenously overlapping with dermal T_{reg} clusters (Fig. 2C) (28). However, despite overall similarities, many genes were differentially expressed between these fractions (Fig. 2D). Although this divergence could, in part, be due to small differences in tissue processing (i.e., the dermal fraction is exposed to an additional enzymatic digestion step), receptor-ligand pairing analysis identified 16 receptor-ligand interactions unique to epithelial T_{reg} s (Fig. 2E). We further stratified the epithelial compartment into hair follicle and IFE cells based on *Krt14*, *Mt2*, *Cd34*, *Ptn*, and *Postn* expression and examined significant interaction pairs (fig. S1D). Most interactions unique to epithelial T_{reg} s were with hair follicle partner cells because T_{reg} s closely associate with the hair follicle epithelium (23, 25). Interactions between *Tgfb1* and *Tgfb1* were enriched with the IFE (Fig. 2E). Consistent with this finding, quantitative reverse transcription polymerase chain reaction (RT-PCR) analysis of purified epithelial T_{reg} s revealed that these cells preferentially expressed *Itgb8* when compared with T_{reg} s purified from the dermis or SDLNs (Fig. 2F). Whereas TGF- β stimulation of dermal and lymph node T_{reg} s induced similar phosphorylation levels of the downstream signaling protein SMAD2/3, epithelial associated T_{reg} s exhibited slightly increased SMAD2/3 phosphorylation (fig. S1E). Together,

these data demonstrate that dermal and epidermal T_{reg} s are largely similar, with both populations preferentially expressing transcriptional pathways relevant to epithelial cell biology.

Activation of latent TGF- β by T_{reg} s delayed epidermal repair

We have previously observed that T_{reg} s facilitate epidermal barrier repair after superficial skin abrasion (29). In this model, repeated application of adhesive tape disrupts the stratum corneum while leaving the underlying tissue layers intact. After this form of injury, T_{reg} s attenuate the C-X-C motif chemokine ligand 5 (CXCL5)/interleukin-17 (IL-17)/neutrophil axis and, in doing so, enable hair follicle stem proliferation and differentiation to help regenerate the epithelium (29). Given these findings and the important role for TGF- β signaling in wound repair (6), we hypothesized that skin T_{reg} s harness the TGF- β pathway to modulate epithelial responses to injury (Fig. 3A). The specific enrichment of *Itgb8* on epithelial associated T_{reg} s led us to postulate that T_{reg} activation of latent TGF- β could regulate epithelial cell activity after injury. Unlike other integrins, $\alpha\beta 8$ does not transmit tensile force through the actin cytoskeleton, and its only known function is activation of latent TGF- β via conformational exposure of TGF- β to TGF- β R (30–32). Murine T_{reg} s in secondary lymphoid organs have

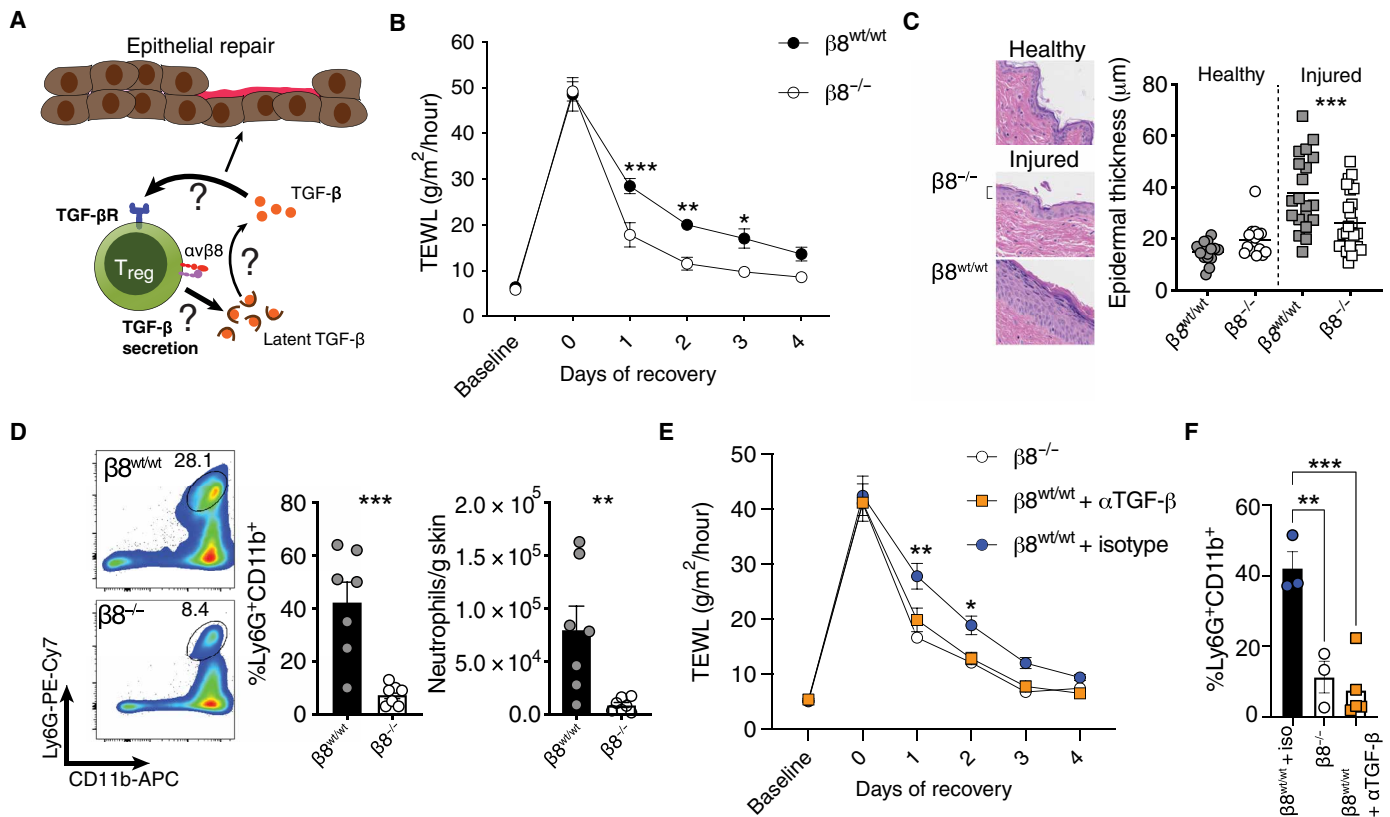


Fig. 3. Activation of latent TGF- β by T_{reg} s delayed epidermal repair. (A) Schematic depicting our hypothesis that skin T_{reg} s harness the TGF- β pathway to modulate epithelial responses to injury. scRNA-seq analysis identified skin T_{reg} s as poised to respond to TGF- β , secrete TGF- β , and activate latent TGF- β . All of these functions could regulate epithelial cell biology. (B) Kinetic transepidermal water loss (TEWL) measurements taken before and after tape stripping *Foxp3*^{Cre-ERT2}*Itgb8*^{fl/fl} mice treated with either vehicle ($\beta 8^{wt/wt}$) or tamoxifen ($\beta 8^{-/-}$). (C) Representative histology and quantification of epidermal thickness comparing $\beta 8^{wt/wt}$ and $\beta 8^{-/-}$ mice before or 2 days after injury. Symbols indicate individual measurements. (D) Flow cytometric quantification of neutrophils in the skin after tape stripping. PE-Cy7, R-phycoerythrin-cyanine7; APC, allophycocyanin. (E and F) T_{reg} $\beta 8^{wt/wt}$ mice were treated intradermally with 100 μ g of TGF- β -neutralizing or isotype control antibody immediately before tape stripping. TEWL (E) and neutrophil accumulation (F) were monitored as before. All data are representative of at least two independent experiments; $n = 3$ to 7 mice per group; error bars are SEMs. Statistical analysis was performed using two-way ANOVA and Sidak's multiple comparisons test (B, E, and F) or an unpaired Student's two-tailed t test (C and D). * $P < 0.5$, ** $P < 0.01$, and *** $P < 0.001$.

been reported to increase expression of $\alpha\beta 8$ after activation, consistent with the activated signature of skin T_{reg} s (Fig. 1B) (14, 15).

To test whether *Itgb8* was involved in T_{reg} responses to epithelial injury, we bred *Foxp3*^{ERT2-Cre} mice (33) to *Itgb8*^{fllox/fllox} mice (34) to generate an inducible and lineage-specific model for deleting $\alpha\beta 8$ in T_{reg} s. Sex-matched *Foxp3*^{ERT2-Cre}*Itgb8*^{fllox/fllox} littermates were treated with tamoxifen or vehicle control and subjected to superficial skin injury by tape stripping the stratum corneum (fig. S2A). Administration of tamoxifen reduced T_{reg} expression of *Itgb8*, although deletion was incomplete (fig. S2B). Skin barrier recovery was quantified by measuring transepidermal water loss and epidermal thickness. Unexpectedly, deletion of $\alpha\beta 8$ on T_{reg} s resulted in significantly faster epithelial barrier repair, with reduced transepidermal water loss early in the recovery process (Fig. 3B). Consistent with this finding, tamoxifen-treated mice also exhibited significantly reduced epidermal thickening (Fig. 3C). These results were recapitulated in *Foxp3*^{Cre}*Itgb8*^{fllox/fllox} animals whose T_{reg} s are constitutively deficient in $\alpha\beta 8$ (fig. S2, C and D).

Epidermal injury induces an inflammatory response that recruits neutrophils into the skin. Excessive influx of these cells hinders epithelial differentiation and delays restoration of the barrier (29). We therefore surveyed immune cell populations in our *Foxp3*^{Cre-ERT2}*Itgb8*^{fllox/fllox} animals (fig. S3A). After tape stripping, tamoxifen-treated animals exhibited notably reduced accumulation of neutrophils in the skin (Fig. 3D). To directly link deletion of $\alpha\beta 8$ on T_{reg} s with TGF- β activity, we treated wild-type mice with TGF- β -neutralizing antibody immediately before tape stripping. In contrast to isotype treatment, TGF- β blockade closely mimicked epidermal repair kinetics and

impaired neutrophil recruitment observed with T_{reg} deletion of *Itgb8* (Fig. 3, E and F). These results support the hypothesis that activation of latent TGF- β by T_{reg} s is a relevant pathway in epithelial repair. Unexpectedly, our data point to a mechanism whereby T_{reg} utilization of this pathway impairs barrier repair capacity instead of facilitating it.

$\alpha\beta 8$ -expressing T_{reg} s promoted innate skin inflammation

Delayed barrier repair, epidermal thickening, and increased neutrophil recruitment are markers of skin inflammation (29). Thus, the delayed epidermal regeneration observed in control animals suggested that the skin of mice with functional integrin $\alpha\beta 8$ on T_{reg} s (i.e., wild-type T_{reg} s) was more inflamed (Fig. 3). Because T_{reg} s efficiently restrain IL-17-mediated inflammation and T_H17 cell differentiation can be induced by TGF- β (29, 35), we examined whether IL-17 production was enhanced in the skin of mice that express functional integrin $\alpha\beta 8$ on T_{reg} s. After barrier injury, there was no difference between wild-type and *Foxp3*^{ERT2-Cre}*Itgb8*^{fllox/fllox} animals in IL-17 production from either $\alpha\beta$ or $\gamma\delta$ T cells, indicating that integrin $\alpha\beta 8$ expression on T_{reg} s played no discernible role in regulating the IL-17 pathway in the skin (Fig. 4, A and B). Similarly, we did not observe any difference in accumulation of T_H1 or T_H2 cells (fig. S3, B and C).

To screen for candidate inflammatory mediators induced by $\alpha\beta 8$ -expressing T_{reg} s, we quantified cytokine and chemokine expression in tape-stripped and uninjured whole skin tissue by quantitative PCR (qPCR) array. Although there were no overt differences at steady state, 2 days after barrier injury, the expression of both *Il24*

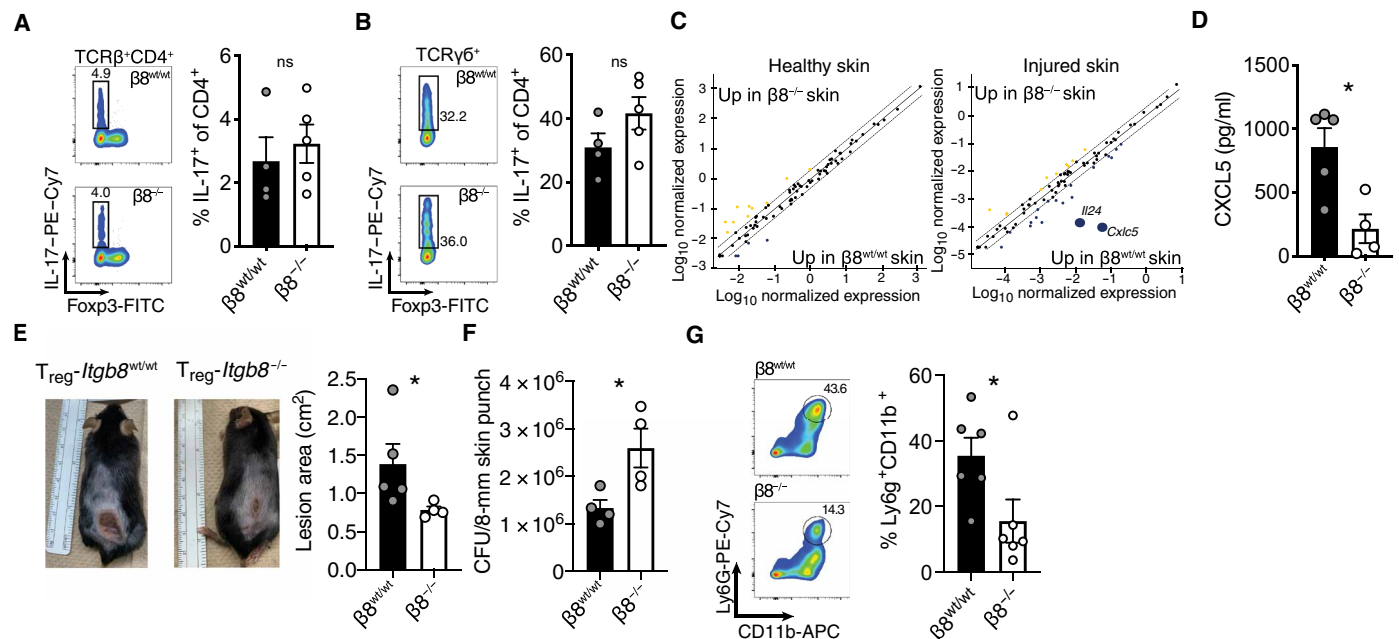


Fig. 4. $\alpha\beta 8$ ⁺ T_{reg} s promoted innate skin inflammation. (A and B) T_{reg} s from *Foxp3*^{Cre-ERT2}*Itgb8*^{fllox/fllox} mice treated with either vehicle or tamoxifen were phenotyped by flow cytometry for IL-17 expression in skin CD4⁺ $\alpha\beta$ T cells (A) or $\gamma\delta$ T cells (B) after epithelial injury. FITC, fluorescein isothiocyanate. (C) Whole skin qPCR array analysis comparing cytokine and chemokine gene expression levels between $\beta 8^{wt/wt}$ and $\beta 8^{-/-}$ healthy or tape strip-injured mice. (D) Whole skin biopsy lysate ELISA for CXCL5. (E and F) T_{reg} $\beta 8^{wt/wt}$ and $\beta 8^{-/-}$ mice intradermally infected with 5×10^7 CFU *S. aureus* (USA300) were monitored for 6 days before 8-mm punch biopsies surrounding lesions were excised for bacterial CFU analysis. (G) Representative flow plots and quantification of neutrophil accumulation in the skin after *S. aureus* infection. Data are representative of at least three independent experiments; $n = 4$ to 7 mice per group; error bars are SEMs. Statistical analysis was performed with an unpaired Student's two tailed t test. * $P < 0.05$.

and *Cxcl5* was markedly reduced in T_{reg} *Itgb8*-deleted mice (Fig. 4C). *Cxcl5* codes for the chemokine CXCL5 and is known to be a potent driver of neutrophil accumulation in tissues (36, 37). We confirmed the reduction in CXCL5 protein levels by enzyme-linked immunosorbent assay (ELISA) on whole skin lysates (Fig. 4D).

Given the proinflammatory function of T_{regs} in promoting neutrophil recruitment, we sought to determine whether the presence or absence of $\alpha\beta8$ on T_{regs} affects their lineage identity and/or stability as regulatory cells. To this end, we quantified transcript levels of T_{reg} lineage-defining genes (*Foxp3*, *Il2ra*, *Ctla4*, *Icos*, and *Irf2*) between *Itgb8*⁺ and *Itgb8*⁻ T_{regs} in our skin scRNA-seq dataset. This analysis revealed no significance differences in T_{reg} -defining genes between these populations (fig. S3D). In addition, we tested whether deletion of *Itgb8* on T_{regs} results in loss of T_{reg} identity and/or stability during steady state. *Foxp3*^{ERT2-Cre}*Itgb8*^{flx/flx} mice treated with tamoxifen proliferated normally (fig. S3E). There were also normal T_{reg} frequencies and absolute numbers in the skin (fig. S3F). T_{regs} in these mice exhibited equivalent protein levels of Foxp3 and CD25 when compared with vehicle-treated controls (fig. S3G). Together, these results demonstrate that bona fide T_{regs} mediated activation of latent TGF- β , which acted to enhance CXCL5 production and neutrophil accumulation acutely after skin barrier injury. Although this comes at a cost of delayed epidermal regeneration, this effect is only transient because barrier recovery in mice with functional integrin $\alpha\beta8$ on T_{regs} is equivalent to that of $\alpha\beta8$ -deficient mice 4 days after injury (Fig. 3B).

Activation of latent TGF- β by T_{regs} protected against bacterial skin infection

A defining function of skin epithelium is to form a tissue barrier to protect against pathogens. After injury, tissue repair processes act to restore barrier function but do so in the context of immediate and ongoing antimicrobial immune responses. We hypothesized that T_{regs} in the skin use the $\alpha\beta8$ /TGF- β pathway to facilitate antimicrobial immunity immediately after epithelial injury. This concept is not without precedent because T_{regs} in the vaginal mucosa help facilitate tissue entry of specific immune cell populations to augment early protective immune responses to viral infections (38). To test this hypothesis, we infected the skin of *Foxp3*^{Cre}*Itgb8*^{flx/flx} or tamoxifen-treated *Foxp3*^{Cre-ERT2}*Itgb8*^{flx/flx} mice with *Staphylococcus aureus* and quantified skin infection area and bacterial burden relative to control animals. Mice with T_{regs} deficient in *Itgb8* exhibited significantly smaller lesions surrounding the injection site, consistent with reduced skin inflammation previously observed in these animals (Fig. 4E). However, these mice maintained about double the bacterial burden, indicating an inability to control the infection (Fig. 4F). Neutrophil infiltration was also reduced (Fig. 4G). Together, these data indicate that although expression of $\alpha\beta8$ on skin T_{regs} slightly reduced the efficiency of epithelial repair, it simultaneously served to enhance innate immunity against bacterial threats occurring immediately after skin barrier breach.

TGF- β signaling did not influence T_{regs} in a cell-intrinsic fashion during epidermal repair

We next set out to elucidate the mechanisms by which T_{reg} -mediated activation of latent TGF- β leads to enhanced innate inflammation after epidermal injury. Because skin T_{regs} are highly enriched in TGF- β pathway genes, including *Tgbr2* and *Tgfb1*, we hypothesized that integrin $\alpha\beta8$ -mediated activation of latent TGF- β by skin T_{regs} works in

an autocrine fashion to control their homeostasis and suppressive capacity in the skin (Fig. 5A). After epidermal injury, there were no differences in the frequency or absolute numbers of T_{regs} in either the skin or SDLNs of *Itgb8*-deleted mice (Fig. 5B and fig. S3H). In addition, flow cytometric quantification of critical T_{reg} effector molecules, such as Foxp3, CD25, and Cytotoxic T-lymphocyte-associated protein 4 (CTLA-4), revealed no differences between these groups (Fig. 5C). Levels of Ki67 were also unchanged, indicating that $\alpha\beta8$ -deficient T_{regs} proliferate normally in vivo after skin barrier disruption (Fig. 5D). These results suggest that activation of latent TGF- β by integrin $\alpha\beta8$ does not markedly alter T_{reg} numbers or activation in the context of skin barrier breach.

To definitively test whether TGF- β signaling acts on T_{regs} in an autocrine fashion to influence their function during epidermal repair, we generated *Foxp3*^{Cre-ERT2}*Tgfb2*^{flx/flx} mice (39). These animals are inducibly refractory to TGF- β signaling due to tamoxifen-sensitive deletion of TGF- β Rii [which is required for productive signaling through all TGF- β Rs (39, 40)], enabling us to determine whether TGF- β acts directly on T_{regs} . Tamoxifen treatment reduced *Tgfb2* transcript and protein levels and impaired pSMAD2/3 induction after TGF- β stimulation (fig. S4, A to C). Deletion of TGF- β signaling in T_{regs} had no effect on skin barrier regeneration because *Foxp3*^{Cre-ERT2}*Tgfb2*^{flx/flx} animals treated with tamoxifen showed no difference from sex-matched littermate, vehicle-treated controls with respect to transepidermal water loss or epidermal thickness (Fig. 5E, F). T_H17 cells also accumulated normally (fig. S4D).

In addition to activating latent TGF- β and responding to this cytokine, T_{regs} can synthesize latent TGF- β (41), and this molecule was highly enriched in skin T_{regs} (Fig. 1E). Thus, we set out to determine whether T_{reg} production of this cytokine plays a role in their early response to epithelial injury (Fig. 5A). To do so, we generated *Foxp3*^{Cre-ERT2}*Tgfb1*^{flx/flx} animals in which latent TGF- β production can be inducibly depleted in T_{regs} (fig. S4, E and F). Upon barrier disruption, mice deficient in T_{reg} production of latent TGF- β had no discernible differences in transepidermal water loss or epidermal thickness when compared with age- and sex-matched littermate controls (Fig. 5, G and H). Together, these results suggest that activation of latent TGF- β and not secretion or cell-intrinsic sensing of this cytokine was the primary mechanism active in skin T_{regs} to promote inflammation after epithelial injury. This raised the possibility that T_{reg} activation of latent TGF- β acts extrinsically on other cell populations in the skin.

Epithelial cell TGF- β sensing promoted neutrophil recruitment during injury

Given that T_{regs} were primed to interact with keratinocytes (Figs. 1, F and G, and 2E), we hypothesized that T_{reg} -mediated activation of TGF- β could act directly on these cells (Fig. 6A). To test this, we generated *K14*^{Cre-ERT2}*Tgfb2*^{flx/flx} animals, which allow lineage-specific inducible deletion of *Tgfb2* in keratinocytes (fig. S4G) (42). We treated age- and sex-matched littermates with tamoxifen or vehicle control, disrupted the epithelial barrier, and quantified epidermal regeneration over time. We observed a significant improvement in transepidermal water loss in mice with a keratinocyte-specific deletion of *Tgfb2*, similar to that observed with *Foxp3*^{Cre}*Itgb8*^{flx/flx} animals (Fig. 6B). ELISA quantification of CXCL5 on whole skin lysates revealed significantly less of this chemokine in mice with keratinocyte-specific deletion of *Tgfb2* (Fig. 6C). Concordantly, both the frequency and absolute number of skin-infiltrating neutrophils were decreased in *Tgfb2*-deficient animals (Fig. 6D). To demonstrate

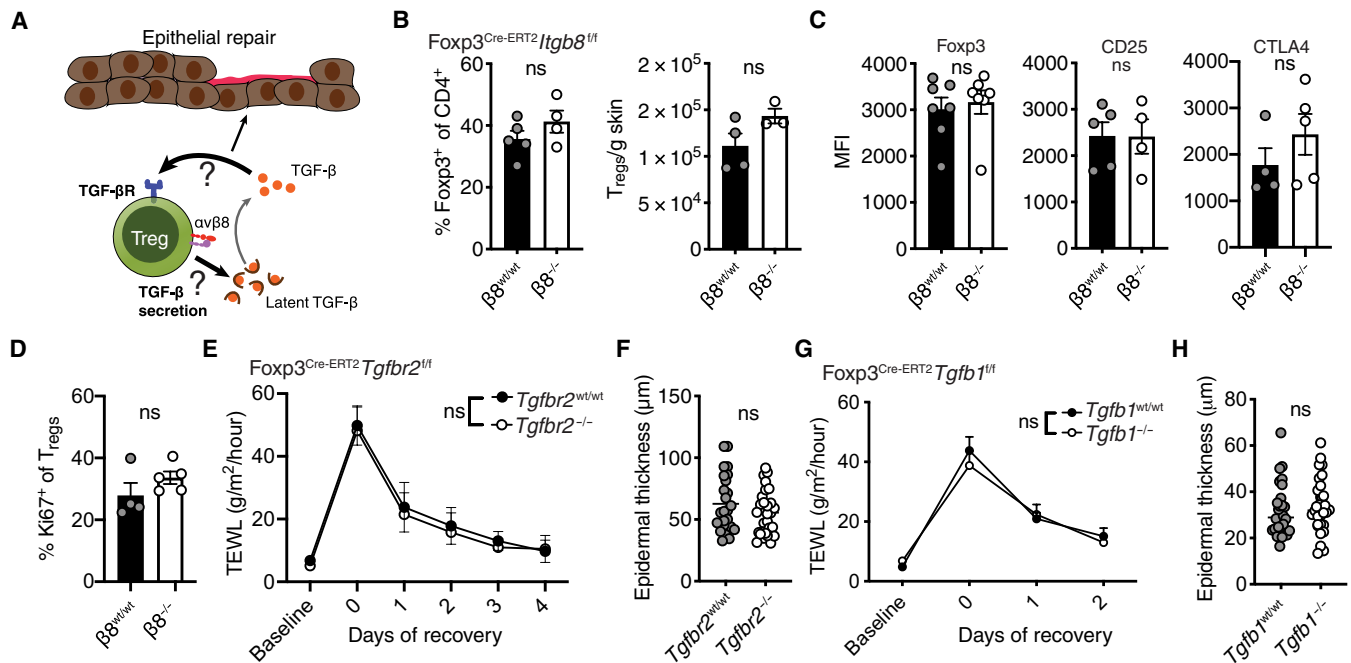


Fig. 5. Cell-intrinsic TGF-β signaling did not drive T_{reg} responses to epidermal injury. (A) Schematic depicting the possibility that skin T_{reg}s either respond to or release TGF-β to regulate epithelial repair. TGF-β secretion could alter tissue bioavailability of this cytokine, whereas TGF-β sensing by T_{reg}s has the potential to alter their functionality. (B to D) T_{reg}s from *Foxp3^{Cre-ERT2}Itgb8^{fl/fl}* mice treated with either vehicle or tamoxifen were phenotyped by flow cytometry for (B) their frequency and absolute abundance in the skin, (C) their expression of lineage-defining proteins, (D) and the frequency of proliferating cells. MFI, median fluorescence intensity. (E) Kinetic transepidermal water loss measurements taken before and after tape stripping *Foxp3^{Cre-ERT2}Tgfb2^{fl/fl}* mice treated with either vehicle or tamoxifen. (F) Epidermal thickness 2 days after injury, as measured from histology. (G) Kinetic transepidermal water loss measurements taken before and after tape stripping *Foxp3^{Cre-ERT2}Tgfb1^{fl/fl}* mice treated with either vehicle or tamoxifen. (H) Epidermal thickness 2 days after injury, as measured from histology. Data are representative of at least two independent experiments; *n* = 4 to 7 per group; error bars are SEMs. Statistical analysis was performed using two-way ANOVA and Sidak's multiple comparisons test (E and G) or an unpaired Student's two-tailed *t* test (B to D, F, and H).

that these effects were due to T_{reg} αvβ8 expression, we examined epithelial TGF-β signaling in T_{reg} *Itgb8*-deleted mice. In *Itgb8*-deficient animals, IFE cells, but not hair follicle-associated cells, had reduced SMAD2/3 phosphorylation. (Fig. 6E). Moreover, IFE cells exhibited significantly less up-regulation of *Cxcl5* expression after tape strip injury (Fig. 6F). Thus, TGF-β signaling in keratinocytes after epidermal injury led to increased CXCL5 production and neutrophil recruitment. Together, our data suggest that T_{reg}s are uniquely positioned and have the molecular machinery to help mediate this effect.

DISCUSSION

T_{reg}s have classically been defined by their ability to robustly control inflammation and promote immune tolerance. Nonetheless, it is increasingly recognized that tissue-resident T_{reg}s influence their local environment in many ways that are seemingly independent of their role in suppressing inflammation (2, 3, 23, 29, 43). Our data are consistent with others (38) and support a model whereby T_{reg}s in barrier tissues enable specific inflammatory pathways acutely after barrier breach. Implicit in this model is that subsets of these cells maintain molecular pathways for instructing tissue biology beyond dampening inflammation. These pathways appear to be driven by tissue specificity and heterogeneity within the T_{reg} pool.

Here, we provide evidence of a transcriptional profile unique to skin T_{reg}s and tailored for specific interaction with cell types comprising their tissue niche. This module was defined by enrichment

of both TGF-β and integrin pathway genes. The genes coding for integrins αv and β8, which belong to both gene sets, were preferentially enriched on skin T_{reg}s and even further on T_{reg}s closely associated with skin epithelium. By examining the transcriptional states of both T_{reg}s and their epithelial counterparts, we were able to find a significant coassociation of these pathways. These interactions were relatively absent in lymph node-derived T_{reg}s. In functionally dissecting the TGF-β molecular circuit in skin T_{reg}s, we revealed that αvβ8-expressing cells activate latent TGF-β to work in a cell-extrinsic fashion to induce keratinocytes to increase expression of the neutrophil chemoattractant CXCL5. Although this process slightly delayed epithelial regeneration after injury, it provided protection from bacterial infection.

Activation of TGF-β by αvβ8 expressed on T_{reg}s suppresses inflammatory T cell responses and attenuates T cell-mediated colitis (14). Our results are consistent with this activity because we have uncovered a locally regulated circuit that underlines the importance of tissue context in the integration of barrier repair and functional immunity. Epithelial cells were the dominant recipient of T_{reg}-activated TGF-β in the context of barrier breach, but we have not ruled out the possibility that effector T cells are also suppressed by this activity. TGF-β simultaneously mediating opposed functions is not unprecedented because this cytokine is able to exert both pro- and anti-inflammatory responses in tissues (7–9). The TGF-β pathway can augment neutrophil accumulation and activation (including through induction of CXCL5), and T_{reg}s themselves are capable of

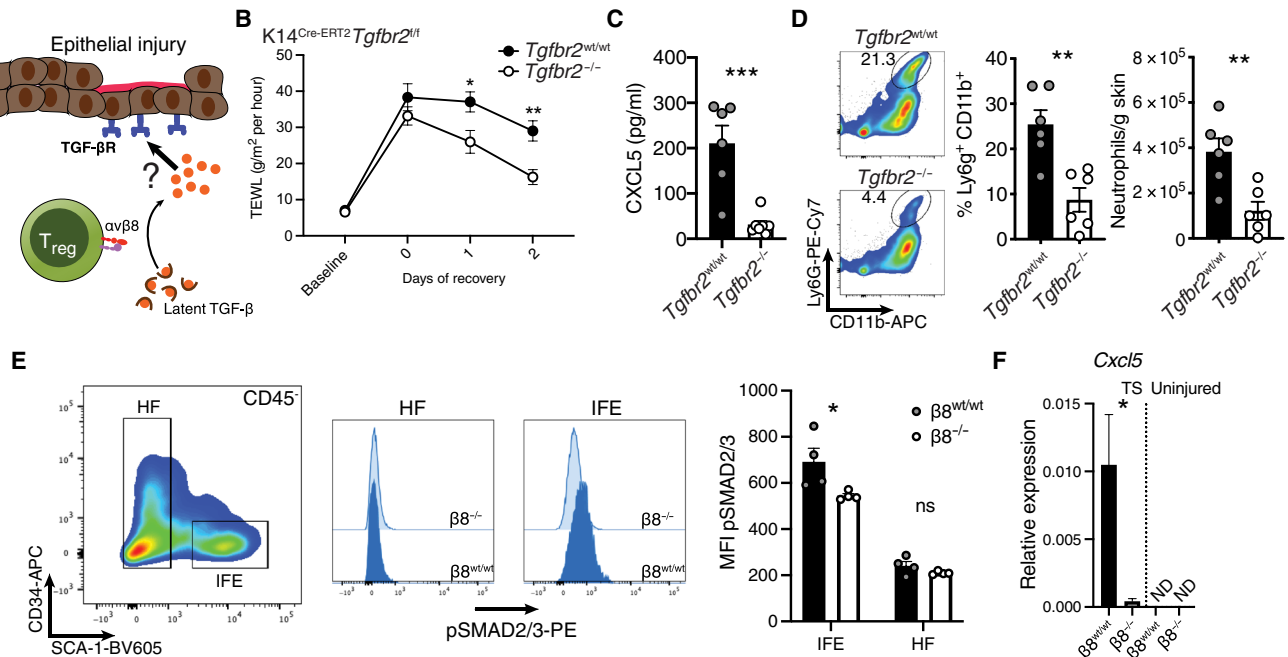


Fig. 6. TGF- β signaling in keratinocytes promoted neutrophil recruitment acutely after epidermal injury. (A) Schematic representation of our hypothesis that during epithelial injury, keratinocytes sense TGF- β activated by T_{reg}s. Although T_{reg}-intrinsic sensing of TGF- β was not important during epithelial repair, a paracrine response through a secondary cell type was possible. (B) Kinetic transepidermal water loss measurements taken before and after tape stripping K14^{Cre-ERT2} *Tgfb2*^{fl/fl} mice treated with either vehicle (*Tgfb2*^{wt/wt}) or tamoxifen (*Tgfb2*^{fl/fl}). (C) Whole skin biopsy lysate ELISA for CXCL5. (D) Flow cytometric quantification of neutrophils in the skin 2 days after injury. (E) Keratinocytes from Foxp3^{Cre-ERT2} *Itgb8*^{fl/fl} mice treated with either vehicle or tamoxifen were phenotyped by flow cytometry for SMAD2/3 phosphorylation after injury. (F) CD45⁺ Sca-1⁺ epidermal cells from T_{reg} β8^{wt/wt} and β8^{-/-} mice were sorted from tape-stripped (TS) or uninjured mice and analyzed for *Cxcl5* expression by qPCR. Data are representative of three independent experiments; *n* = 4 to 6 mice per group; error bars are SEMs. Statistical analysis was performed using two-way ANOVA and Sidak's multiple comparisons test (B) or an unpaired two-tailed *t* test (C to F). ND, not detected.

producing neutrophil-recruiting chemokines (44–47). Expression of αvβ8 on skin T_{reg}s delayed epithelial regeneration but did not prevent it. In addition, activation of latent TGF- β and not secretion of this cytokine by T_{reg}s predominated in this mechanism, suggesting that already available TGF- β was necessary to mediate this acute response. Expression of another TGF- β -activating integrin, αvβ6, is functionally important in the IFE (48). Skin T_{reg}s did not express appreciable amounts of αvβ6. However, it is possible that αvβ6 expression on nonimmune cells synergizes with T_{reg}s to activate latent TGF- β during epidermal injury. A limitation of our study is that we have not dissected how and whether αvβ8 and αvβ6 expression on other cell types contribute to epidermal barrier repair. Although we demonstrated deletion of *Tgfb2* and *Tgfb1* T_{reg}s at both mRNA and protein levels, we were unable to verify protein knockout of αvβ8 with currently available reagents. In addition, deletion of all three genes was not completely efficient in T_{reg}s. Given that tamoxifen-induced deletion of *Itgb8* was incomplete, it is likely that our results underreported the magnitude and physiological significance of this pathway.

Seemingly paradoxical to the current study, our previous work demonstrated that T_{reg}s promote skin healing by suppressing the CXCL5-neutrophil axis (29). However, several notable features of our collective data indicate that these results are not inconsistent. Rather, together, they reveal a critical role for skin T_{reg}s in maintaining tissue integrity by establishing a balance between the opposing need for protective immunity and tissue repair. Ablation of all T_{reg}s

leads to robust skin inflammation and hinders epithelial regeneration, but the potency of this approach obscures more nuanced activity of individual molecular pathways. Data presented herein does not suggest that T_{reg}s lose their suppressive capabilities when expressing αvβ8. Rather, acquisition of this molecular pathway in the context of epithelial injury allowed rapid promotion of neutrophilic infiltration through additional independent rather than opposed mechanisms. This is supported by our observations that skin T_{reg}s maintain normal suppression of IL-17 production from effector T cell populations. Thus, deletion of αvβ8 on T_{reg}s uncoupled T_H17 immunity and neutrophil recruitment in the context of epidermal injury. We postulate that co-option of inflammatory pathways by epithelial associated T_{reg}s may have arisen from evolutionary pressure to balance the kinetics of tissue healing with defense against pathogens. In this context, tissue T_{reg}s may serve as a fulcrum balancing the need for epithelial regeneration with that of acute host defense.

It is becoming increasingly accepted that T_{reg}s residing in non-lymphoid organs are functionally heterogeneous across tissues and distinct from their lymphoid counterparts. Our data support this notion. The fact that these cells promote optimal skin function by both suppressing aberrant inflammation and selectively recruiting inflammatory cells raises a number of interesting avenues for further investigation. In particular, several skin diseases, such as atopic dermatitis, are characterized by uncontrolled inflammation, barrier disruption, and recurrent bacterial or viral infection (49). Patients with congenital T_{reg} defects develop this skin disease (50). Thus,

functional manipulation of specific pathways active in subsets of T_{regs} may be an attractive approach to more precisely restore the balance between inflammation and barrier integrity in patients with atopic dermatitis and other conditions characterized by compromised barrier integrity and chronic inflammation.

MATERIALS AND METHODS

Study design

The objective of this study was to define the transcriptional identity and functional activity of T_{regs} present in the skin. The TGF- β pathway was of specific interest. To investigate this, we surveyed the transcriptional landscape of T_{regs} and epithelial cells isolated from murine skin with scRNA-seq. Functional investigations were carried out with inducible Cre/Lox mouse model systems. Replicate information and animal numbers are indicated in appropriate figure legends. Quantification of histology was performed blinded.

Mice

Foxp3^{GFP} [B6.Cg-*Foxp3*^{tm2(EGFP)*Tch*}/J], *Foxp3*^{Cre} [B6.129(CG)-*Foxp3*^{tm4(YFP/cre)*Ayr*}/J], *Foxp3*^{Cre-ERT2} [*Foxp3*^{tm9(EGFP/cre/ERT2)*Ayr*}/J], and *K14*^{Cre-ERT2} [Tg(KRT14-cre/ERT)20Efu/J] mice were purchased from the Jackson Laboratory. *Itgb8*^{fllox/fllox} (B6.129-*Itgb8*^{tm2Lfr}) animals were a gift of D. Sheppard [University of California at San Francisco (UCSF)]. *Tgfbri*^{fllox/fllox} (B6;129-*Tgfbri*^{tm1Karl}/J) animals were gifted by Q. Tang (UCSF), and *Tgfb1*^{fllox/fllox} (C57BL/6J-*Tgfb1*^{em2Lutzy}/Mmjax) mice were gifted by M. Krummel (UCSF). Cre lines were crossed with LoxP strains until homozygosity to generate animals conducive for inducible and lineage-specific gene deletion. Cre-ERT2 animals were treated for five consecutive days with 2.5 mg of tamoxifen dissolved in corn oil intraperitoneally (Sigma-Aldrich) to induce Cre recombinase activity. All animal experiments were performed on littermate age- and sex-matched 6- to 16-week-old mice maintained through routine breeding at the UCSF School of Medicine in a specific pathogen-free facility. Experimental procedures were approved by Institutional Animal Care and Use Committee and performed in accordance with guidelines established by the Laboratory Animal Resource Center at UCSF.

Skin injury, bacterial infection, and TGF- β blocking

Mechanical injury of mouse skin and transepidermal water loss were measured as previously described (29). Briefly, the entire back skin was shaved and tape stripped 5 to 10 times using cellophane tape for three consecutive days. Transepidermal water loss was measured with a Tewameter TM 300 probe (Courage + Khazaka electronic GmbH) according to the manufacturer's protocols. Baseline (before tape stripping), day 0 (immediately after last tape strip), and recovery (every 24 hours thereafter) measurements were collected on each quadrant of the back skin to monitor epidermal healing. For bacterial injection mice were challenged intradermally with 5×10^7 CFU (colony-forming units) of *S. aureus* strain USA300. Bacterial burden was measured on day 6 by excising an 8-mm punch of the skin surrounding the lesion injection site, homogenizing tissue with a TissueRuptor (QIAGEN), and plating supernatants in a dilution series on chloramphenicol containing agar petri dishes. Colonies were manually counted after an overnight incubation. TGF- β blocking experiments were conducted by a single intradermal injection of 100 μ g of clone 1D11.16.8 per mouse (Thermo Fisher Scientific) or isotype control MOPC-21 (BioXCell) immediately before the

first round of tape stripping. Mice were sedated with isoflurane during shaving, mechanical injury, and transepidermal water loss measurements.

Tissue processing

Single-cell suspensions of the skin were prepared by finely mincing tissues and digesting for 45 min in an enzyme cocktail containing collagenase XI, deoxyribonuclease, and hyaluronidase in complete RPMI 1640 at 37°C. For separation of epithelial and dermal fractions, skin sections were floated on thermolysin-containing buffer, as previously described (27). Lymphoid tissues were mashed through a 40- μ m filter to release leukocytes. ELISA analysis was performed by homogenizing whole skin punch biopsies and lysing cells with Cell Lysis Buffer II (R&D Systems). CXCL5 was detected with the Mouse LIX DuoSet ELISA Kit (R&D Systems). For qPCR analysis of whole skin, RNA was extracted using the RNeasy Fibrous Tissue Kit (QIAGEN) and the RT² Profiler PCR Array Kit (PAMM-150ZC, QIAGEN) used for complementary DNA synthesis and amplification. *Itgb8*, *Tgfb2*, *Tgfb1*, and *Cxcl5* expression on sorted cells was quantified using the TaqMan Gene Expression Cells-to-CT Kit (Thermo Fisher Scientific) and TaqMan assays Mm00623991_m1, Mm03024075_m1, Mm03024091_m1, Mm00436451_g1, and Mm01178820_m1.

For histological analysis, skin tissue was fixed in 10% formalin and paraffin-embedded. Tissue was stained with hematoxylin and eosin (H&E) by the UCSF Mouse Pathology Core. Epidermal thickness was quantified from H&E slides in a blinded manner by measuring five randomly selected points per section using Zen 2 (Zeiss) software analysis tools. Immunofluorescent tissue staining was performed by fixing skin from *Foxp3*^{GFP} reporter mice with 2% paraformaldehyde for 4 hours, washed with phosphate-buffered saline (PBS), and left in 30% sucrose overnight before embedding in optimal cutting temperature compound and freezing. Slides were mounted with 4',6-diamidino-2-phenylindole (DAPI)-containing medium before imaging on a standard fluorescent microscope.

Flow cytometry and cell sorting

Cells prepared as described in the above tissue processing section were stained with Ghost 510 viability dye (Tonbo Biosciences) in PBS. Surface makers were stained in PBS with 2% fetal bovine serum. Cells were fixed and permeabilized with the *Foxp3*/Transcription Factor Staining Buffer Set (eBioscience) before staining intracellular antigens. For cytokine analysis, cells were stimulated for 4 hours with phorbol ester and ionomycin (Cell Stimulation Cocktail, Tonbo Biosciences) before staining. For flow cytometric analysis, samples were run on an LSRFortessa analyzer (355-, 405-, 488-, 532-, 561-, and 640-nm laser configuration; BD Biosciences) in the UCSF flow cytometry core and collected using FACSDiva software (BD Biosciences). Compensation was performed using UltraComp eBeads as single-color controls (Thermo Fisher Scientific). Data were analyzed using FlowJo software (Tree Star Inc.). Fluorescence-activated cell sorting isolation of T_{regs} was performed using a FACSARIA Fusion sorter (BD Biosciences). *Foxp3*^{GFP} reporter mice were used to allow identification of CD45⁺TCR- β ⁺CD4⁺*Foxp3*⁺ cells.

Fluorophore-conjugated antibodies specific for mouse and human antigens were purchased from eBioscience, BD Biosciences, R&D Systems, and BioLegend. Antibodies for staining mouse cells are as follows: α -CD8 α (clone 53-6.7), α -TCR- β (clone H57-597); α -CD4 (clone GK1.5), TCR- γ (clone GL3), α -CD45 (clone 30-F-11), α -Ki67 (clone B56), α -Ly6G (clone 1A8), α -CD11b (clone M1/70),

α -Foxp3 (clone FJK-16s); α -CTLA-4 (clone UC10-4F10-11), α -CD25 (clone PC61.5), α -CD34 (clone MEC14.7), α -IL-17 (TC11-18H10.1), α -IL-5 (TRFK5), α -IFN- γ (XMGI.2), α -SCA-1 (clone D7), α -pSMAD2/3 (clone 072-670), α -TGF- β R2 (polyclonal), and α -LAP (TW7-20B9).

RNA sequencing

scRNA-seq was performed by the UCSF Functional Genomics core facility using the 10x Chromium Single Cell 3' Gene Expression Kit, according to the manufacturer's instructions (10x Genomics, Pleasanton, CA). Paired-end sequencing (150) was performed on a NovaSeq 6000 instrument. The Cell Ranger analysis pipelines (version 3.0.2, 10x Genomics) were then used to process the generated sequencing data. The R package Seurat (version 3.0) was used for gene expression analysis. Filtered gene-barcode matrices were loaded, and quality control steps were performed (low quality or dying cells and cell doublets/multiplets were excluded from subsequent analysis). Data were normalized and scaled, and then, linear dimensional reduction using principle components analysis was performed. Highly variable genes were used to perform unsupervised clustering, and nonlinear dimensional reduction with uniform manifold approximation and projection (UMAP) was used to visualize the data. Differential genes were calculated with Seurat on nonintegrated data. CellPhoneDB analysis was performed as previously described (22). Bulk RNA sequencing was performed by Expression Analysis, Quintiles (Morrisville, NC). Normalized RNA counts were determined using the R package DESeq2.

Statistical analysis

Statistical analyses were performed with Prism software (GraphPad). A two-tailed unpaired Student's *t* test or two way analysis of variance (ANOVA) with Sidak's multiple comparisons test was applied to calculate *P* values assuming a normal sample distribution, as indicated. The test used for each analysis is indicated in the figure legends. All experiments were performed with at least two independent trials, as indicated. Statistical analysis of scRNA-seq data was handled with either Seurat (28), GSEA (19), PANTHER (18), or CellPhoneDB (22), as indicated. *P* values correlate with symbols as follows: ns, not significant; **P* < 0.05; ***P* < 0.01; ****P* < 0.001; and *****P* < 0.0001.

SUPPLEMENTARY MATERIALS

immunology.sciencemag.org/cgi/content/full/6/62/eabg2329/DC1

Figs. S1 to S4

Table S1

[View/request a protocol for this paper from Bio-protocol.](#)

REFERENCES AND NOTES

- P. P. Lui, I. Cho, N. Ali, Tissue regulatory T cells. *Immunology* **161**, 4–17 (2020).
- M. Panduro, C. Benoist, D. Mathis, Tissue Tregs. *Annu. Rev. Immunol.* **34**, 609–633 (2016).
- M. Ito, K. Komai, S. Mise-Omata, M. Iizuka-Koga, Y. Noguchi, T. Kondo, R. Sakai, K. Matsuo, T. Nakayama, O. Yoshie, H. Nakatsukasa, S. Chikuma, T. Shichita, A. Yoshimura, Brain regulatory T cells suppress astrogliosis and potentiate neurological recovery. *Nature* **565**, 246–250 (2019).
- R. J. Miragaia, T. Gomes, A. Chomka, L. Jardine, A. Riedel, A. N. Hegazy, N. Whibley, A. Tucci, X. Chen, I. Lindeman, G. Emerton, T. Krausgruber, J. Shields, M. Haniiffa, F. Powrie, S. A. Teichmann, Single-cell transcriptomics of regulatory T cells reveals trajectories of tissue adaptation. *Immunity* **50**, 493–504.e7 (2019).
- M. Delacher, C. D. Imbusch, A. Hotz-Wagenblatt, J.-P. Mallm, K. Bauer, M. Simon, D. Riegel, A. F. Rendeiro, S. Bittner, L. Sanderink, A. Pant, L. Schmidleithner, K. L. Brabant, B. Echtenachter, A. Fischer, V. Giunchiglia, P. Hoffmann, M. Edinger, C. Bock, M. Rehli, B. Brors, C. Schmidl, M. Feuerer, Precursors for nonlymphoid-tissue Treg cells reside in secondary lymphoid organs and are programmed by the transcription factor BATF. *Immunity* **52**, 295–312.e11 (2020).
- M. Pakyari, A. Farrokhi, M. K. Maharlooie, A. Ghahary, Critical role of transforming growth factor β in different phases of wound healing. *Adv. Wound Care* **2**, 215–224 (2013).
- G. Han, C. A. Williams, K. Salter, P. J. Garl, A. G. Li, X. J. Wang, A role for TGF β signaling in the pathogenesis of psoriasis. *J. Invest. Dermatol.* **130**, 371–377 (2010).
- M. Lodyga, B. Hinz, TGF- β 1—A truly transforming growth factor in fibrosis and immunity. *Semin. Cell Dev. Biol.* **101**, 123–139 (2020).
- M. A. Travis, D. Sheppard, TGF- β activation and function in immunity. *Annu. Rev. Immunol.* **32**, 51–82 (2014).
- J. E. Konkel, D. Zhang, P. Zanvit, C. Chia, T. Zangar-Murray, W. Jin, S. Wang, W. J. Chen, Transforming growth factor- β signaling in regulatory T cells controls T helper-17 cells and tissue-specific immune responses. *Immunity* **46**, 660–674 (2017).
- M. O. Li, Y. Y. Wan, R. A. Flavell, T cell-produced transforming growth factor- β 1 controls T cell tolerance and regulates Th1- and Th17-cell differentiation. *Immunity* **26**, 579–591 (2007).
- W. Ouyang, O. Beckett, Q. Ma, M. O. Li, Transforming growth factor- β signaling curbs thymic negative selection promoting regulatory T cell development. *Immunity* **32**, 642–653 (2010).
- S. Z. Josefowicz, L.-F. Lu, A. Y. Rudensky, Regulatory T cells: Mechanisms of differentiation and function. *Annu. Rev. Immunol.* **30**, 531–564 (2012).
- J. J. Worthington, A. Kelly, C. Smedley, D. Bauché, S. Campbell, J. C. Marie, M. A. Travis, Integrin $\alpha\beta$ -mediated TGF- β activation by effector regulatory T cells is essential for suppression of T-cell-mediated inflammation. *Immunity* **42**, 903–915 (2015).
- J. P. Edwards, A. M. Thornton, E. M. Shevach, Release of active TGF- β 1 from the latent TGF- β 1/GARP complex on T regulatory cells is mediated by integrin β_8 . *J. Immunol.* **193**, 2843–2849 (2014).
- C. Ferreira, L. Barros, M. Baptista, B. Blankenhans, A. Barros, P. Figueiredo-Campos, S. Konjar, A. Lainé, N. Kamenjarin, A. Stojanovic, A. Cerwenka, H. C. Probst, J. C. Marie, M. Veldhoen, Type 1 Treg cells promote the generation of CD8+ tissue-resident memory T cells. *Nat. Immunol.* **21**, 766–776 (2020).
- D. Haribhai, W. Lin, L. M. Relland, N. Truong, C. B. Williams, T. A. Chatila, Regulatory T cells dynamically control the primary immune response to foreign antigen. *J. Immunol.* **178**, 2961–2972 (2007).
- P. D. Thomas, M. J. Campbell, A. Kejarawal, H. Mi, B. Karlak, R. Davenport, K. Diemer, A. Muruganujan, A. Narechania, PANTHER: A library of protein families and subfamilies indexed by function. *Genome Res.* **13**, 2129–2141 (2003).
- A. Subramanian, P. Tamayo, V. K. Mootha, S. Mukherjee, B. L. Ebert, M. A. Gillette, A. Paulovich, K. S. Pomeroy, T. R. Golub, E. S. Lander, J. P. Mesirov, Gene set enrichment analysis: A knowledge-based approach for interpreting genome-wide expression profiles. *Proc. Natl. Acad. Sci. U.S.A.* **102**, 15545–15550 (2005).
- D. Mu, S. Cambier, L. Fjellbirkeland, J. L. Baron, J. S. Munger, H. Kawakatsu, D. Sheppard, V. C. Broadus, S. L. Nishimura, The integrin $\alpha\beta$ mediates epithelial homeostasis through MT1-MMP-dependent activation of TGF- β 1. *J. Cell Biol.* **157**, 493–507 (2002).
- S. Joost, K. Annusver, T. Jacob, X. Sun, T. Dalessandri, U. Sivan, I. Sequeira, R. Sandberg, M. Kasper, The molecular anatomy of mouse skin during hair growth and rest. *Cell Stem Cell* **26**, 441–457.e7 (2020).
- M. Efremova, M. Vento-Tormo, S. A. Teichmann, R. Vento-Tormo, CellPhoneDB: Inferring cell–cell communication from combined expression of multi-subunit ligand–receptor complexes. *Nat. Protoc.* **15**, 1484–1506 (2020).
- N. Ali, B. Zirik, R. S. Rodriguez, M. L. Pauli, H. A. Truong, K. Lai, R. Ahn, K. Corbin, M. M. Lowe, T. C. Scharschmidt, K. Taravati, M. R. Tan, R. R. Ricardo-Gonzalez, A. Nosbaum, M. Bertolini, W. Liao, F. O. Nestle, R. Paus, G. Cotsarelis, A. K. Abbas, M. D. Rosenblum, Regulatory T cells in skin facilitate epithelial stem cell differentiation. *Cell* **169**, 1119–1129.e11 (2017).
- D. Burzyn, W. Kuswanto, D. Kolodin, J. L. Shadrach, M. Cerletti, Y. Jang, E. Sefik, T. G. Tan, A. J. Wagers, C. Benoist, D. Mathis, A special population of regulatory T cells potentiates muscle repair. *Cell* **155**, 1282–1295 (2013).
- I. K. Gratz, H. A. Truong, S. H. Y. Yang, M. M. Maurano, K. Lee, A. K. Abbas, M. D. Rosenblum, Cutting edge: Memory regulatory T cells require IL-7 and Not IL-2 for their maintenance in peripheral tissues. *J. Immunol.* **190**, 4483–4487 (2013).
- F. Thelen, D. A. Witherden, Get in touch with dendritic epithelial T cells! *Front. Immunol.* **11**, 1656 (2020).
- C. Walzer, M. Benathan, E. Frenk, Thermolysin treatment: A new method for dermo-epidermal separation. *J. Invest. Dermatol.* **92**, 78–81 (1989).
- T. Stuart, A. Butler, P. Hoffman, C. Hafemeister, E. Papalexi, W. M. Mauck III, Y. Hao, M. Stoeckius, P. Smibert, R. Satija, Comprehensive integration of single-cell data. *Cell* **177**, 1888–1902.e21 (2019).
- A. N. Mathur, B. Zirik, I. C. Boothby, M. Tan, J. N. Cohen, T. M. Mauro, P. Mehta, M. M. Lowe, A. K. Abbas, N. Ali, M. D. Rosenblum, Treg-cell control of a CXCL5-IL-17 inflammatory axis promotes hair-follicle-stem-cell differentiation during skin-barrier repair. *Immunity* **50**, 655–667.e4 (2019).

30. M. G. Campbell, A. Cormier, S. Ito, R. I. Seed, A. J. Bondesson, J. Lou, J. D. Marks, J. L. Baron, Y. Cheng, S. L. Nishimura, Cryo-EM reveals integrin-mediated TGF- β activation without release from latent TGF- β . *Cell* **180**, 490–501.e16 (2020).
31. J. Wang, Y. Su, R. E. Jacob, J. R. Engen, T. A. Springer, General structural features that regulate integrin affinity revealed by atypical $\alpha\text{V}\beta 8$. *Nat. Commun.* **10**, 5481 (2019).
32. A. Ozawa, Y. Sato, T. Imabayashi, T. Uemura, J. Takagi, K. Sekiguchi, Molecular basis of the ligand binding specificity of $\alpha\text{V}\beta 8$ integrin. *J. Biol. Chem.* **291**, 11551–11565 (2016).
33. Y. P. Rubtsov, R. E. Niec, S. Josefowicz, L. Li, J. Darce, D. Mathis, C. Benoist, A. Y. Rudensky, Stability of the regulatory T cell lineage in vivo. *Science* **329**, 1667–1671 (2010).
34. J. M. Proctor, K. Zang, D. Wang, K. Wang, L. F. Reichardt, Vascular development of the brain requires $\beta 8$ integrin expression in the neuroepithelium. *J. Neurosci.* **25**, 9940–9948 (2005).
35. M. Pawlak, A. W. Ho, V. K. Kuchroo, Cytokines and transcription factors in the differentiation of CD4⁺ T helper cell subsets and induction of tissue inflammation and autoimmunity. *Curr. Opin. Immunol.* **67**, 57–67 (2020).
36. J. Mei, Y. Liu, N. Dai, M. Favara, T. Greene, S. Jeyaseelan, M. Poncz, J. S. Lee, G. S. Worthen, CXCL5 regulates chemokine scavenging and pulmonary host defense to bacterial infection. *Immunity* **33**, 106–117 (2010).
37. J. Mei, Y. Liu, N. Dai, C. Hoffmann, K. M. Hudock, P. Zhang, S. H. Guttentag, J. K. Kolls, P. M. Oliver, F. D. Bushman, G. S. Worthen, Cxcr2 and Cxcl5 regulate the IL-17/G-CSF axis and neutrophil homeostasis in mice. *J. Clin. Invest.* **122**, 974–986 (2012).
38. J. M. Lund, L. Hsing, T. T. Pham, A. Y. Rudensky, Coordination of early protective immunity to viral infection by regulatory T cells. *Science* **320**, 1220–1224 (2008).
39. P. Levéen, J. Larsson, M. Ehinger, C. M. Cilio, M. Sundler, L. J. Sjöstrand, R. Holmdahl, S. Karlsson, Induced disruption of the transforming growth factor β type II receptor gene in mice causes a lethal inflammatory disorder that is transplantable. *Blood* **100**, 560–568 (2002).
40. R. Derynck, E. H. Budi, Specificity, versatility, and control of TGF- β family signaling. *Sci. Signal.* **12**, eaav5183 (2019).
41. K. Nakamura, A. Kitani, W. Strober, Cell contact-dependent immunosuppression by CD4⁺CD25⁺ regulatory T cells is mediated by cell surface-bound transforming growth factor β . *J. Exp. Med.* **194**, 629–644 (2001).
42. V. Vasioukhin, L. Degenstein, B. Wise, E. Fuchs, The magical touch: Genome targeting in epidermal stem cells induced by tamoxifen application to mouse skin. *Proc. Natl. Acad. Sci. U.S.A.* **96**, 8551–8556 (1999).
43. L. A. Kalekar, J. N. Cohen, N. Prevel, P. M. Sandoval, A. N. Mathur, J. M. Moreau, M. M. Lowe, A. Nosbaum, P. J. Wolters, A. Haemel, F. Boin, M. D. Rosenblum, Regulatory T cells in skin are uniquely poised to suppress profibrotic immune responses. *Sci. Immunol.* **4**, eaaw2910 (2019).
44. T. Parekh, B. Saxena, J. Reibman, B. N. Cronstein, L. I. Gold, Neutrophil chemotaxis in response to TGF- β isoforms (TGF- β 1, TGF- β 2, TGF- β 3) is mediated by fibronectin. *J. Immunol.* **152**, 2456–2466 (1994).
45. C. Haider, J. Hnat, R. Wagner, H. Huber, G. Timelthaler, M. Grubinger, C. Coulouarn, W. Schreiner, K. Schlagen, W. Sieghart, M. Peck-Radosavljevic, W. Mikulits, Transforming growth factor- β and Axl induce CXCL5 and neutrophil recruitment in hepatocellular carcinoma. *Hepatology* **69**, 222–236 (2019).
46. M. E. Himmel, S. Q. Crome, S. Ivson, C. Piccirillo, T. S. Steiner, M. K. Levings, Human CD4⁺FOXP3⁺ regulatory T cells produce CXCL8 and recruit neutrophils. *Eur. J. Immunol.* **41**, 306–312 (2011).
47. R. A. Fava, N. J. Olsen, A. E. Postlethwaite, K. N. Broadley, J. M. Davidson, L. B. Nanney, C. Lucas, A. S. Townes, Transforming growth factor beta 1 (TGF- β 1) induced neutrophil recruitment to synovial tissues: Implications for TGF- β -driven synovial inflammation and hyperplasia. *J. Exp. Med.* **173**, 1121–1132 (1991).
48. J. Mohammed, L. K. Beura, A. Bobr, B. Astry, B. Chicoine, S. W. Kashem, N. E. Welty, B. Z. Igyártó, S. Wijeyesinghe, E. A. Thompson, C. Matte, L. Bartholin, A. Kaplan, D. Sheppard, A. G. Bridges, W. D. Shlomchik, D. Masopust, D. H. Kaplan, Stromal cells control the epithelial residence of DCs and memory T cells by regulated activation of TGF- β . *Nat. Immunol.* **17**, 414–421 (2016).
49. P. Y. Ong, D. Y. M. Leung, Bacterial and viral infections in atopic dermatitis: A comprehensive review. *Clin Rev Allergy Immunol* **51**, 329–337 (2016).
50. F. Barzaghi, L. Passerini, R. Bacchetta, Immune dysregulation, polyendocrinopathy, enteropathy, X-linked syndrome: A paradigm of immunodeficiency with autoimmunity. *Front. Immunol.* **3**, 211 (2012).

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